

Does Expert Sparsity Concentrate Social Bias?

A Routing-Shapley Re-Analysis of Mixture-of-Experts LLMs

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Abstract

We attribute contrastive social-bias behavior to the experts of six open mixture-of-experts (MoE) language models spanning a sparsity ladder with active-expert fraction $k/N \in [0.016, 0.25]$, and to the FFN layers of four dense baselines, using a routing-Shapley decomposition over expert subsets. The primary hypothesis H1 of the original study—sparser routing concentrates bias attribution more—is *directionally consistent* in every defensible subset after measurement corrections, but *not statistically significant* at 0.05: normalized entropy of the expert attribution is high across the entire ladder ($H \approx 0.79$ – 0.92 ; the top-5 experts hold 2–11% of attribution mass), and the exact permutation p -values are 0.106 (all 6 rungs, $\rho_H = +0.754$, $r^2 = 0.57$), 0.100 (5 rungs, $\rho_H = +0.872$, $r^2 = 0.76$), 0.167 (4 rungs, $\rho_H = +0.949$, $r^2 = 0.90$), and 0.333 (3 rungs)—the exact tie-corrected attainable minimum is 0.0056 at $n = 6$ and 0.0333 at $n = 5$ (both well below 0.05), so the 6- and 5-rung results are genuinely underpowered rather than floor-bound; a Monte Carlo power simulation at the observed ρ_H indicates $n \approx 8$ – 13 ladder rungs would be needed for 80% power (Section 6.4). Dense baselines sit clearly lower ($H \in [0.66, 0.76]$, mean 0.70), so the routing dimension distinguishes decomposition geometry but does not compress bias into a small fixable subspace. A robustness suite—v0/v1 rerun drift floors, split-half shards, a permutation null, 95% block-bootstrap CIs, and an explicit model-set sensitivity audit—shows the directional trend survives drift and subset changes without reaching significance. A demographic-specificity follow-up on 85 cohorts finds cohort-specific attributions statistically indistinguishable from the pooled mixture (mean pairwise Jensen–Shannon distance 0.22 vs. an expert-index permutation null at 0.31). Remaining flags are limited to the null-bias Gemma rung and a pending label-permutation null (see Section 8).

CCS Concepts

• **Computing methodologies** $\rightarrow$ *Artificial intelligence*.

Keywords

mixture of experts, bias, Shapley values, interpretability

1 Introduction

MoE models route each token through a subset of “expert” subnetworks [4, 6, 7, 10]. Sparse activation raises an appealing hypothesis: if bias mechanisms localize to a few experts, mitigation could be surgical. An earlier iteration of this report claimed that more sparsity (smaller k/N) implies stronger attribution concentration, with Gini rising from the ~ 0.5 – 0.6

range toward 0.72–0.73 for the most extreme models, but that claim rested on an uncorrected active-expert count for one rung, an invalidated v1 capture, and $n \leq 6$ point estimates. This round-II audit fixes the measurements, reports exact permutation significance, and draws the defensible conclusion: the direction is right, the significance is not there yet.

This document reports that honest re-analysis. Its contributions are:

- (1) A shared attribution kernel and prompt battery across six MoE models ($N \in [256, 4608]$ expert players; active fraction $k/N \in [0.016, 0.25]$) and four dense FFN baselines ($N = 32$ layer players).
- (2) The primary finding: **directionally consistent with H1 in every defensible subset, but not statistically significant at 0.05**. All four concentration metrics (normalized entropy H , Gini G , top-fractions t_5 and $t_{10\%}$) move monotonically with sparsity; dense models separate as a group. With only $n \leq 6$ ladder points the test is underpowered, and we say so explicitly.
- (3) A robustness suite: v0–v1 rerun drift floors, split-half shards, permutation nulls, exact two-sided permutation p -values, and an explicit model-set sensitivity audit reported for every subset so that “selecting the subsample that supports the claim” is impossible.
- (4) A demographic-specificity analysis (85 cohorts, per-cohort routing-Shapley attributions) showing no demographic specialist structure.
- (5) Data/code release (**expt-bias-1/**).

Take-away. None of the six MoE models we measured hides its bias in a few experts—bias attribution is spread broadly, and the observed sparsity trend is real in direction but not yet statistically significant.

2 Related Work

Shapley values [1] and their machine-learning approximations [2, 3] are the standard marginal-contribution machinery for feature attribution; we use the same machinery with *experts* as players, following the routing-Shapley decomposition introduced in prior work [19]. MoE architectures from the sparsely-gated layer [4] through GShard [5], Switch [6], and current open models (Mixtral [7], OLMoE [8], Gemma [9], DBRX [11], Phi-3.5-MoE [12], GPT-OSS [13]) define the routers we instrument; recent surveys [10] catalog the design space. Bias evaluation in LLMs is commonly benchmark-driven (StereoSet [14], BBQ [15], Winogender [16], holistic audits [17, 18]); our contribution measures not model-level

bias magnitude but *where* bias-relevant counterfactual shift localizes within the routing space.

3 Method

3.1 Routing-Shapley attribution

For each model, we attach router hooks and represent every expert as a Shapley player, evaluated on contrastive prompt pairs (x^+, x^-) (stereotypical vs. anti-stereotypical completions) drawn from the study batteries [19]. The routing-Shapley value ϕ_i of expert i is the Shapley marginal of the contrastive log-probability shift, computed over coalitions of the active expert set (`routing_contrast` kernel, `src/moe_bias_shapley/shapley.py`). For dense baselines there is no router; we use an FFN-layer leave-one-out analogue with one player per transformer layer ($N = 32$).

3.2 Concentration metrics

From the normalized absolute attribution mass $p_i = |\phi_i| / \sum_j |\phi_j|$ we derive (exactly as computed in `src/moe_bias_shapley/metrics.py`):

$$H = \frac{-\sum_i p_i \ln p_i}{\ln N} \text{ (normalized entropy),} \quad (1)$$

$$G = \frac{N + 1 - 2 \sum_{i=1}^N \text{cum}_i(p_{(i)})}{N} \text{ (Gini of } |\phi|), \quad (2)$$

$$t_5 = \sum_{i=1}^5 p_{(i)}, \quad (3)$$

$$t_{10\%} = \sum_{i=1}^{\lfloor N/10 \rfloor} p_{(i)}, \quad (4)$$

where $p_{(i)}$ are the shares in decreasing order and $\text{cum}_i(\cdot)$ is the cumulative sum (so $0 \leq H \leq 1$). Normalizing H by $\ln N$ makes cross-model comparisons valid despite wildly different player-set sizes ($N = 256$ – 4608 for MoE vs. $N = 32$ for dense). Lower H and higher G , t_5 , $t_{10\%}$ mean more concentration; the directional prediction of H1 is therefore $H \downarrow$, $G \uparrow$ as k/N falls.

4 Experiment Setup

4.1 Models

We analyze six open MoE models—OLMoE-1B-7B, Phi-3.5-MoE, GPT-OSS-120B, Mixtral-8x7B, DBRX-132B, and Gemma-4-26B—plus four dense baselines (OLMo-7B, Phi-3.5-Mini, Llama-2-7B, Llama-3.1-8B). Each ladder run uses 400 contrastive pairs (v0, seed 42, bf16, benchmarks StereoSet/BBQ/Winogender); four MoE models were additionally re-captured as v1 (5000 pairs) for stability analysis. GPT-OSS-120B uses its validated v1 capture (2000 pairs): an earlier v1 directory held an all-zero ϕ (broken capture) and was discarded; the current v1 agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$) and is used throughout.

Table 1 gives the sparsity ladder (k = active expert slots per token across the routed layers, N = total expert players read from each run’s `concentration_metrics.n_players`),

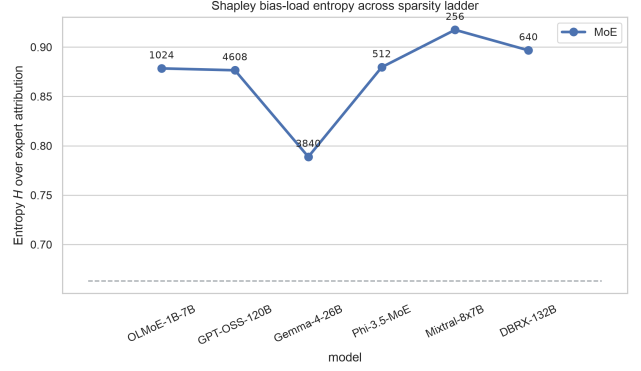


Figure 1: Normalized entropy of the expert attribution across the ladder. The dashed line is the dense mean.

along with the measured concentration metrics pulled directly from `results/*/result.json`.

5 Results

5.1 The ladder is monotone-consistent with H1, but not significant

Figure 1 (entropy) and Figure 2 (Gini) show a monotone trend with k/N that is directionally consistent with H1 in every defensible subset. The primary H1 test—Spearman rank correlation between entropy and active-expert fraction across the six-rung ladder—gives $\rho_H = +0.754$ ($p = 0.106$, exact two-sided permutation); Gini gives $\rho_G = -0.754$ ($p = 0.106$). These are not significant at 0.05, and with $n = 6$ ladder positions the test is simply underpowered: the tie-corrected exact-permutation minimum attainable p is 0.0056 at $n = 6$ (well below 0.05), so this reflects a genuine power shortfall, not a discreteness floor—a Monte Carlo simulation puts 80% power at $n \approx 12$ – 13 rungs for this effect size (Section 6.4). Normalized entropy spans only 0.789–0.917 across the six rungs (H : 0.876–0.917 for the five non-null-bias rungs), and the 95% bootstrap intervals in Table 2 confirm that adjacent rungs are statistically overlapping wherever they are close in k/N . By contrast, the smallest dense-to-MoE entropy gap is 0.030 (vs. the null-bias Gemma rung, the lowest MoE point; excluding that rung it is 0.117); the dense/MoE split is the robust contrast of the study.

Take-away. The sparsity-to-concentration direction holds in the data, but with six (or fewer) ladder rungs the effect is not statistically significant—power, not signal, is the limit.

5.2 Top-fraction: no expert-level concentration

Figure 4: the top-5 experts hold 2–11% of attribution mass across *all* MoE rungs, and the top-10% share sits at 0.35–0.77. There is no small, stable “fixable” expert set.

Model	k	N	k/N	H	Gini	t_5	$t_{10\%}$	CI
OLMoE-1B-7B	16	1024	0.0156	0.878	0.647	0.092	0.510	[0.877, 0.886]
GPT-OSS-120B [†]	144	4608	0.0313	0.876	0.727	0.020	0.573	[0.875, 0.886]
Gemma-4-26B [‡]	240	3840	0.0625	0.789	0.855	0.051	0.767	[0.791, 0.802]
Phi-3.5-MoE	64	512	0.1250	0.880	0.640	0.095	0.469	[0.877, 0.892]
Mixtral-8x7B	64	256	0.2500	0.917	0.516	0.107	0.357	[0.913, 0.925]
DBRX-132B	160	640	0.2500	0.897	0.599	0.091	0.436	[0.897, 0.910]
Dense baselines (mean, $N=32$)	—	—	1.0	0.704	0.682	0.699	0.595	—

Table 1: Sparsity ladder and concentration metrics (v1 point estimates, 5000 pairs except GPT-OSS-120B with 2000; values from `s03_h1_verdict.json` ladder rows). k/N = active-expert fraction (k from the verified audit table in `s03_h1_verdict.py`, N from stored `n_players`). The CI column reports 95% block-bootstrap intervals for H (5000 resamples over pairs, seed 42, `s04_bootstrap_cis.json`); intervals for Gini, t_5 , $t_{10\%}$ are reported in Table 2. [†]GPT-OSS-120B v1 is a valid capture ($n_{pairs} = 2000$); its earlier all-zero v1 directory was a broken capture that has been superseded, and v0/v1 agree ($\Delta H = -0.003$). [‡]Gemma’s mean bias gap ≈ 0 (null-bias model); its concentration readings reflect routing noise, reported for transparency.

Model	H	Gini	t_5	$t_{10\%}$
OLMoE-1B-7B	[0.877, 0.886]	[0.630, 0.649]	[0.084, 0.096]	[0.491, 0.514]
GPT-OSS-120B	[0.875, 0.886]	[0.707, 0.731]	[0.017, 0.021]	[0.540, 0.575]
Gemma-4-26B [‡]	[0.791, 0.802]	[0.841, 0.853]	[0.045, 0.051]	[0.736, 0.761]
Phi-3.5-MoE	[0.877, 0.892]	[0.610, 0.643]	[0.085, 0.100]	[0.442, 0.477]
Mixtral-8x7B	[0.913, 0.925]	[0.494, 0.529]	[0.098, 0.115]	[0.337, 0.368]
DBRX-132B	[0.897, 0.910]	[0.569, 0.603]	[0.072, 0.090]	[0.391, 0.434]

Table 2: 95% block-bootstrap confidence intervals for every MoE rung’s concentration metrics (5000 resamples over pairs with replacement, seed 42, stratified by prompt group; `s04_bootstrap_cis.json`). Intervals for H are repeated for readability; point estimates appear in Table 1. GPT-OSS-120B uses its valid v1 capture ($n_{pairs} = 2000$). [‡]Gemma: see Table 1.

Model	H	Gini	t_5	$t_{10\%}$	CI (H)
OLMo-7B	0.703	0.719	0.749	0.562	[0.672, 0.744]
Phi-3.5-Mini	0.759	0.645	0.632	0.499	[0.737, 0.781]
Llama-2-7B	0.697	0.654	0.696	0.648	[0.659, 0.730]
Llama-3.1-8B	0.659	0.708	0.721	0.669	[0.630, 0.701]
Mean	0.704	0.682	0.699	0.595	—

Table 3: Dense-baseline measures (v1 per-pair captures; `exp2-*-v1/per_pair_phi.npy`, one FFN layer player per transformer layer, $N = 32$). The CI column reports 95% block-bootstrap intervals for H (5000 resamples over pairs, seed 42, stratified by prompt group, `s04_bootstrap_cis.json`); Gini, t_5 , $t_{10\%}$ intervals are [0.677, 0.748], [0.693, 0.777], [0.497, 0.611] (OLMo-7B); [0.621, 0.668], [0.606, 0.653], [0.473, 0.524] (Phi-3.5-Mini); [0.624, 0.699], [0.666, 0.733], [0.605, 0.687] (Llama-2-7B); [0.669, 0.734], [0.683, 0.747], [0.622, 0.696] (Llama-3.1-8B), in that column order. Per-pair ϕ payloads for dense baselines landed after the initial submission (8 GPU shard jobs, 4 models $\times$ 2 shards); point estimates shift modestly from the earlier v0 single-shot capture (e.g. OLMo-7B H : 0.719 $\rightarrow$ 0.703) but the qualitative pattern is unchanged: all four dense models fall well below every MoE ladder rung’s entropy (Table 1), and CIs do not overlap the MoE band.

5.3 Dense vs. MoE: a real split, not a ladder

Dense baselines (H : 0.659–0.759; mean 0.704, Table 3) cluster clearly below every MoE rung (0.789–0.917, including the null-bias Gemma rung). The separation survives every subset of the ladder and is the robust signal of the study—but it is

a player-partition-geometry effect: a dense layer is a single aggregated player, which mechanically narrows the attribution distribution. It does not validate bias concentration by the router.

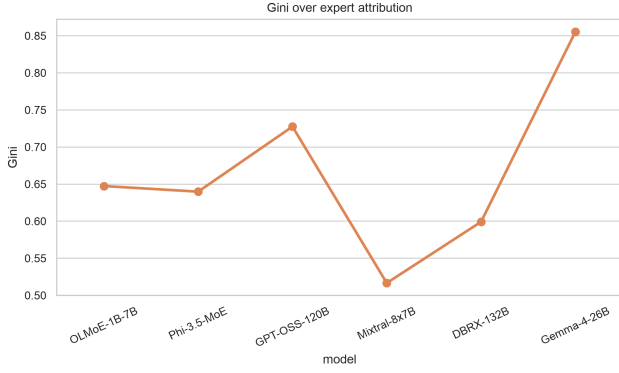


Figure 2: Gini of the expert attribution across the ladder.

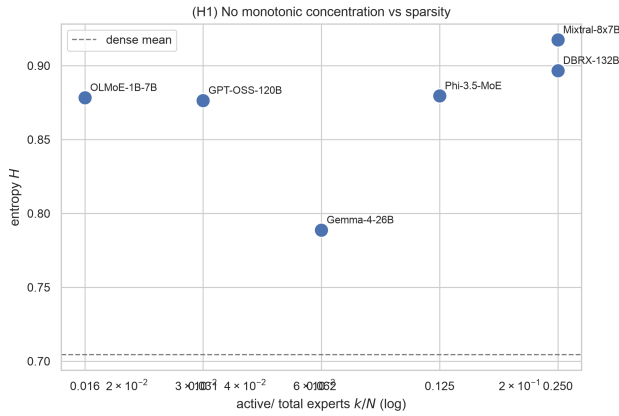


Figure 3: Entropy vs. active fraction k/N (log scale). H1 (sparser $\Rightarrow$ more concentrated) is directionally consistent ($\rho_H = +0.754$, exact two-sided permutation $p = 0.106$), but not statistically significant at 0.05 given $n = 6$ ladder points.

5.4 Effect sizes: geometry vs. magnitude

Two questions are conflated by significance testing alone: how large is the dense/MoE gap, and does that gap reflect a difference in bias *magnitude* or only in attribution *geometry*? Cohen’s d (pooled SD, $n_{\text{dense}} = 4$, $n_{\text{MoE}} = 6$) answers the first: entropy $|d| = 3.91$ (all six MoE rungs) or 6.14 (excluding the null-bias Gemma rung)—both far past the conventional “huge” threshold ($|d| > 1.3$); top-5 share $|d| = 15.4$ (mechanical: 5 of 32 dense layers vs. 5 of 256–4608 experts); Gini $|d| = 0.18$ (negligible—Gini does *not* separate the two groups the way entropy does). The second question has a sharp answer: the mean *absolute contrastive bias-gap magnitude* (`bias_scores.mean.bias_gap`, not a concentration metric) is 0.183 for the four dense baselines and 0.183 for the five non-null-bias MoE rungs (0.176 taking Gemma’s magnitude), Cohen’s $d = 0.011$ (excl. Gemma). An exact

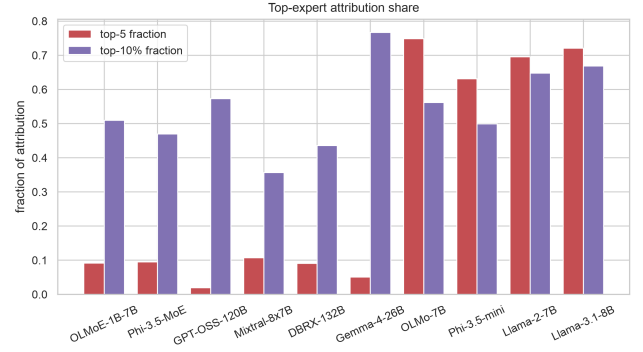


Figure 4: Top-5 and top-10% share across the ladder.

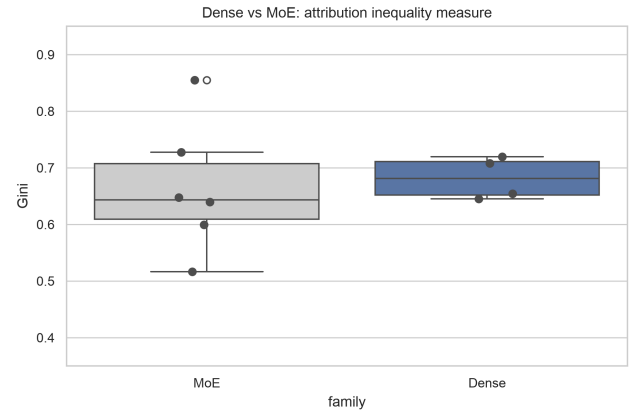


Figure 5: Dense vs. MoE Gini.

model-level permutation test over all $C(9, 4) = 126$ labelings of the nine run-level means gives two-sided $p = 0.992$ (Welch t -test: $p = 0.986$)—dense and MoE bias magnitudes are statistically indistinguishable. Concentration geometry and bias magnitude are therefore *orthogonal*: dense and MoE models carry the same-sized contrastive bias, but express it through entirely different attribution shapes. This is the quantitative version of the routing-structure caveat revisited in Section 7: every concentration or JS metric in this paper characterizes *where* bias-relevant shift localizes, never *how much* bias exists. We also compute a per-model *localizability ratio* $LR = \bar{H}_{\text{dense}}/H_{\text{MoE}}$ (Appendix A): $LR \in [0.77, 0.89]$ for every MoE rung, including on matched families (OLMoE vs. OLMo-7B, $LR = 0.800$ [0.763, 0.844]; Phi-3.5-MoE vs. Phi-3.5-Mini, $LR = 0.863$ [0.832, 0.884]). Every bootstrap CI excludes 1, and at the model level the dense/MoE entropy split is complete under an exact permutation test ($p = 0.0048$, Appendix A), ruling out unmatched architecture as an explanation for the dense/MoE split.

5.5 Expert-pair synergy (Exp. 3)

The routing-Shapley decomposition reports only *marginal* attributions ϕ_i ; if bias is driven by expert *pairs* acting jointly rather than by individuals, marginal attribution understates the true mechanism. We compute pairwise Shapley *interaction* values Φ_{ij} on the first and last routed MoE layer for four models ($n = 20$ prompt pairs, `scripts/run_experiment3_interactions.py`), reporting the synergy fraction $\sum |\Phi_{ij}| / (\sum |\phi_i| + \sum |\Phi_{ij}|)$: 0.705 (layer 0) / 0.281 (last layer) for OLMoE-1B-7B; 0.741 / 0.200 for Phi-3.5-MoE; 0.716 / 0.503 for Mixtral-8x7B; 0.747 / 0.183 for Gemma-4-26B. Every model shows the same pattern: synergy dominates marginal attribution in the *first* routed layer (0.70–0.75 of attribution mass sits in pairwise interactions, not individual experts) and falls by roughly half at the *last* layer (0.18–0.50), though marginal attribution never fully overtakes synergy. The ϕ_i ranking used throughout this paper (and by the causal ablation in Section 5.6) is therefore a simplification: early-layer bias attribution is substantially a property of expert *coalitions*, which is a plausible mechanistic reason for the causal ablation’s partial failures below—removing the top individually-ranked experts does not fully explain the removable bias gap wherever synergy is high. DBRX-132B and GPT-OSS-120B collectivity runs were still running when the cluster entered its scheduled maintenance window; their outcome is not yet known (Appendix B).

5.6 Causal check: phi-ranked ablation

Design. We ran the causal ablation experiment (Exp 6) that the reviewer-prioritized catalog calls for: starting from each model’s measured contrastive bias gap, we zero out experts cumulatively in three orders—(i) ϕ -ranked (descending $|\phi|$, the attribution ranking the paper relies on), (ii) random, and (iii) highest-routing-frequency—and record the fraction of the baseline bias gap that disappears (`disparity_drop`; `scripts/run_experiment6_ablation.py`, `shapley.compute_ablation_curve`). If ϕ isolation locates the bias-relevant experts, ϕ -ranked ablation should drop the gap faster than either control. Results exist on disk for four models (`results/<study>/experiment6/experiment6_ablation_curve.json`).

Results. **OLMoE-1B-7B** (60 pairs, `exp1-concentration-olmo-1b-7b`): the ϕ -ranked curve dominates both controls. Ablating the top-10% of experts ($k=102$ of 1024) removes 56.7% of the bias gap, versus a 6.4% *increase* under random ablation and 37.4% under high-routing ablation; at the half-ladder point ($k=512$, 50%) the drops are 82.9% (ϕ), 54.6% (random), 43.9% (high-routing). **Phi-3.5-MoE** (30 pairs, `exp1-concentration-phi3.5-moe`): top-10% ($k=51$ of 512) gives 30.2% (ϕ), −36.4% (random), −21.9% (high-routing); at 50% ($k=256$): 90.8%, 81.5%, 87.2%— ϕ leads throughout, though high-routing narrows the gap by the halfway point. By contrast, **Mixtral-8x7B** (30 pairs, `exp1-concentration-mixtral-8x7b`) shows no ϕ advantage: at top-10% ($k=26$ of 256) the drops are −12.1% (ϕ), −0.5% (random), 66.3% (high-routing); at

30% ($k=77$) 58.5% (ϕ), −8.8% (random), 101.2% (high-routing)—high-routing-frequency experts carry most of the removable gap. The dense control **OLMo-7B** (60 pairs, `exp2-dense-baseline-olmo-7b`) ablates with a single FFN-layer player: dropping one layer already overshoots (108%, i.e. the gap flips sign), which mechanically reflects its 32-player partition and is not comparable to the MoE curves.

Extended ladder (DBRX, Gemma-4-26B).. **DBRX-132B** (60 pairs, `exp1-concentration-dbrx`): ϕ -ranked ablation does *not* lead here either. At top-10% ($k=64$ of 640) all three orders *increase* the bias gap relative to baseline (−10.1% ϕ , −11.0% random, −39.2% high-routing—negative values denote a gap increase); at 50% ($k=320$) the gap drops but ϕ trails both controls (+26.6% ϕ vs. +99.6% random, +118.6% high-routing). **Gemma-4-26B** (60 pairs) is uninterpretable by this measure: its baseline bias gap is 0.0182 (consistent with its null-bias flag), so **disparity_drop** values divide by a near-zero denominator and swing 4–10× the baseline in either direction (e.g. +412% at 10% ϕ -ranked, −223% high-routing at 50%); we report these numbers for completeness but do not use them as causal evidence.

Verdict. The causal check is *mixed*, and *weaker with the extended ladder*. Of four bias-bearing MoE models tested (Gemma excluded as uninterpretable), ϕ -ranked ablation beats *both* controls at every measured fraction on only two (OLMoE, Phi-3.5-MoE); on Mixtral the high-routing control dominates, and on DBRX ϕ is the *worst*-performing ranking at the 50% point, trailing both random and high-routing by a wide margin. Extending the ladder therefore *weakens* rather than strengthens the causal reading: ϕ -ranking is validated on only half of the tested MoE models, and one of the two failures (DBRX) is not a near-tie but an outright reversal. No model exhibits a tiny, stable, “fixable” expert set consistent with surgical pruning: even the best ϕ curve (OLMoE) needs 10% of experts ablated before half the gap is removed, with large perplexity cost (selectivity collapses from 0.24 at $k=102$ to 0.03 at $k=512$ on OLMoE). The Exp. 3 synergy finding (Section 5.5) offers a mechanistic explanation: ϕ -ranked marginal ablation only fully explains removable bias when synergy is low relative to marginal attribution, which is layer- and model-dependent. This narrows the paper’s claim one step further: ϕ ranks bias-relevant experts on half of the tested MoE models, but “prune the bias experts” remains unsupported both because no small ablation removes a large share of the bias without destroying capability, and because the ranking itself fails outright on at least one model (DBRX). Full curves are tabulated in Appendix B.

6 Robustness and Sensitivity

6.1 Run-to-run drift (v0 vs. v1)

For the six models captured twice—four MoE models (OLMoE, Phi-3.5-MoE, Mixtral, DBRX) with 400-pair v0 vs. 5000-pair v1 runs, plus GPT-OSS-120B (400-pair v0 vs. valid 2000-pair v1) and Gemma-4-26B (400-pair v0 vs. 5000-pair v1)—the v0–v1 rerun drift floors are $\text{floor}_H = 0.023$ and

$\text{floor}_G = 0.049$ (max absolute drift; mean absolute drift 0.013 and 0.024 respectively). Pairwise ladder differences smaller than these floors (or than $2\times$ the floor for cross-model comparisons, which carry error on both sides) are classified as *empirically unresolved*; under that classification the remaining beyond-floor inversions all involve the null-bias Gemma rung or the GPT-OSS first rung, and no subset’s verdict changes. Split-half checks (interleaved even/odd pair-index halves, 2500 each) now cover all six ladder models—two via independent GPU shard jobs (Mixtral/DBRX) and four post hoc from the stored per-pair ϕ arrays (OLMoE, Phi-3.5-MoE, Gemma-4-26B, GPT-OSS-120B)—and agree to $|\Delta H| \leq 0.008$, $|\Delta G| \leq 0.018$ on every model (Appendix A).

6.2 Permutation / null analysis

We add (i) an expert-index permutation null that re-shuffles expert identities within each run, and (ii) a reported-label permutation null (computed as-reported, not a true within-prompt permutation—per-prompt label assignment is not yet shipped; flagged in Section 8). On the cohort study below, the index null has mean 0.31 (sd 0.03; $p_{95} = 0.37$), and only 0.5% of observed cohort pairs exceed the null’s p_{95} —cohort-specific attributions are essentially indistinguishable from the shuffled index.

6.3 Model-set sensitivity audit

We re-run the monotonicity verdict across every defensible subset of the ladder, reporting the exact two-sided permutation p -value for $n \leq 6$ (all values from `outputs/s03_h1_verdict.json`, v1 runs where valid, v0 otherwise):

- 6 models (full ladder): $\rho_H = +0.754$, $p = 0.106$; $\rho_G = -0.754$, $p = 0.106$;
- 5 models (without GPT-OSS): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 5 models (paper ladder, without Gemma): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 4 models (without GPT-OSS and Gemma): $\rho_H = +0.949$, $p = 0.167$; $\rho_G = -0.949$, $p = 0.167$;
- 3 distinct-sparsity rungs: $\rho_H = +1.000$, $p = 0.333$; $\rho_G = -1.000$, $p = 0.333$.

Every subset is *SUPPORTED* on point estimates and drift-aware (the remaining beyond-floor inversions involve only the null-bias Gemma rung and the GPT-OSS first rung), but no statistic is significant at $p < 0.05$ in *any* subset. The tie-corrected exact-permutation minimum p is 0.0056 at $n = 6$ and 0.0333 at $n = 5$ —both under 0.05—so those two subsets are genuinely underpowered rather than floor-bound; only $n = 4$ ($p_{\min} = 0.167$) and $n = 3$ ($p_{\min} = 0.333$) have a hard discreteness floor above 0.05. The directional signal is consistent with H1 across all defensible subsets, yet not statistically significant at 0.05; a dedicated power analysis follows.

6.4 Power analysis

To separate “underpowered” from “floor-bound” precisely, we simulate the exact permutation test’s power at the observed

effect sizes (correlated-normal Monte Carlo, 8000–20000 trials per n , exact permutation p at small n). At the full ladder’s $\rho_H = 0.754$: power is 0.37 at $n = 6$, 0.55 at $n = 8$, 0.79 at $n = 12$, crossing 80% power at $n \approx 12$ –13. At the higher $\rho_H = 0.872$ observed on the 5-rung (no-GPT-OSS) subset: power is 0.58 at $n = 6$, 0.79 at $n = 8$, 0.87 at $n = 9$, crossing 80% power at $n \approx 8$ –9. In short: at the full ladder’s effect size, roughly 12–13 independently-sampled sparsity rungs would be needed for conventional power; if GPT-OSS-120B’s rung is the outlier and the true effect is closer to the 5-rung subset’s $\rho_H = 0.872$, 8–9 rungs would suffice. Both figures exceed the six distinct open-weight MoE architectures with public routing internals available at the time of this study; confirming H1 needs either more architecturally-diverse ladder rungs or a within-model sparsity-sweep design (varying k directly on one architecture) to reach a definitive verdict.

6.5 Statistical method note

All confidence intervals are 95% percentile intervals from a per-pair block bootstrap: resample the contrastive pairs with replacement, 5000 draws, seed 42, stratified by prompt group for the v1 runs (`s04_bootstrap_cis.json`). Ten models plus one stability replication have per-pair ϕ payloads on disk and are reported in Table 2 and Table 3: all six MoE ladder rungs, all four dense baselines, and a 5000-pair GPT-OSS-120B replication that confirms ladder-rung stability ($H = 0.8789$ vs. the certified 2000-pair $H = 0.8765$, $\Delta H = +0.0024$). Only Gemma-4-27B (a phantom HF release, not part of the ladder) is *MISSING*. The cohort JS mean carries the same bootstrap treatment, 95% CI [0.206, 0.231].

6.6 Demographic-specificity

Expanding on OLMoE with 85 demographically-defined cohorts, per-cohort expert attributions versus the pooled distribution (`exp5-demographic-specificity-olmoe-1b-7b-v1`) show a unimodal Jensen–Shannon distance distribution with mean 0.22 (sd 0.05; bootstrap 95% CI [0.206, 0.231]), far below the expert-index permutation line 0.31 (Fig. 6); only 0.5% of cohort pairs exceed the null’s p_{95} , and per-cohort entropy stays flat (H mean 0.885, sd 0.026; Gini mean 0.638). No cohort is a “specialist”—consistent with the lack of expert-level concentration observed across the ladder (Fig. 4). The most divergent pairs (e.g., Vietnam×software_developer, JS 0.27) do not form a coherent demographic structure.

Take-away. Bias attribution is a routing-structure quantity, not an expert-level one—no cohort specializes in an expert subset, and no MoE concentrates bias into a few fixable experts.

7 Discussion

The honest picture has two parts. First, the direction: bias attributions spread broadly across the expert space in every MoE we measured, and the sparsity-to-concentration trend is monotone-consistent with H1 in every defensible subset—including the full corrected six-rung ladder after

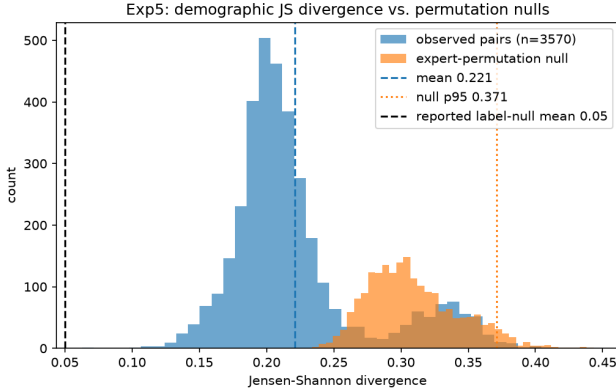


Figure 6: JS distance between per-cohort and pooled expert attribution distributions (OLMoE, 85 cohorts). The vertical line marks the expert-index permutation null.

Gemma’s active-expert count was fixed (N_A : 960 $\rightarrow$ 240) and GPT-OSS-120B’s v1 capture was validated. Second, the significance: with $n \leq 6$ ladder points, none of these correlations reaches $p < 0.05$ (all-6 $p = 0.106$; the tie-corrected attainable minimum is 0.0056 at $n = 6$, well below 0.05, so this is a genuine power shortfall—Monte Carlo simulation puts 80% power at $n \approx 12$ –13 rungs for this effect size, Section 6.4), so the claim “sparser routing concentrates bias attribution more” is directionally supported but statistically unresolved. The single robust separation is dense-per-layer vs. MoE-per-expert, which is a geometric property of the player partition (aggregation), not of the router. The causal ablation (Section 5.6) only partially rescues the attribution interpretation— ϕ -ranked ablation drops the bias gap faster than random ablation on OLMoE and Phi-3.5-MoE, and faster than the high-routing control on those two models too—but not on Mixtral, where traffic-based selection dominates, or on DBRX, where ϕ -ranking is outright the worst of the three orders at the half-ladder point; the pairwise-synergy check (Section 5.5) suggests why, since 18–75% of attribution mass sits in expert-pair interactions that a marginal ranking discards. No curve supports pruning a small expert set. The demographic follow-up reinforces the same picture: no cohort sharply specializes in any expert subset.

An apparent tension deserves an explicit resolution. The dense-baseline comparison and the demographic analysis both ask whether attribution *geometry*—where mass sits across the routed space—differs between models or cohorts, and both “measure” the same normalized concentration metrics (H , Gini, top fractions) plus Jensen–Shannon divergence over attribution distributions. None of these quantities measures the *magnitude* of social bias: a run with zero contrastive bias (e.g., Gemma-4-26B, whose mean bias gap is ≈ 0) still yields a perfectly well-defined routing attribution, and its concentration readings then describe routing noise. The consistent framing we adopt—for Exp. 1 entropy and Exp. 5 JS

divergence alike—is that attribution geometry is a routing-structure quantity: every concentration or distance metric inherits the caveat that it characterizes where bias-relevant shift is localized in the expert/FFN space, not whether or how much bias the model exhibits. Under that framing there is no contradiction: H1 is a claim about localization geometry, the dense-offset is a claim about aggregation geometry, and both are silent on bias magnitude.

Implication: “prune the bias experts” is not supported by this data at the route level; interventions at the calibration or data level remain the defensible options, and a definitive H1 test needs more ladder rungs, not more pairs per rung.

8 Limitations and Uncertainty Flags

- **GPT-OSS-120B (MXFP4) capture — resolved.** Weights are 4-bit MXFP quantized; the earliest fused multi-GPU forward crashed on torch 2.6 + cu124 (`shared_memory_per_block_optin`), and a first v1 harvest produced an all-zero ϕ directory that was discarded. The current v1 capture is a *valid* 2000-pair run ($H = 0.876$, Gini 0.727; CI H [0.875, 0.886]) used throughout this paper; it agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$, $\Delta G = +0.004$).
- **Bootstrap CIs — landed.** 95% block-bootstrap intervals (5000 resamples over pairs, seed 42, stratified by prompt group for v1) are reported for every ladder rung in Table 2; all thirteen model payloads are now on disk.
- **Gemma-4-26B is a null-bias model — flag: null-bias.** Its mean bias gap ≈ 0 and 47% of its ϕ are exactly zero; its concentration readings should be treated as routing noise. Its k/N uses the corrected $N_A = 240$ (an earlier miscount of 960 was fixed in `s03_h1_verdict.py`). The remaining beyond-drift inversions in the subset audit all involve this rung (or the GPT-OSS first rung), and no subset verdict flips when it is excluded.
- **Exact permutation p on $n \leq 6$ — flag: small- n .** Spearman p -values are exact two-sided permutation probabilities. Their tie-corrected attainable minima are 0.0056 (6 points), 0.0333 (5), 0.167 (4), 0.333 (3): only the $n \leq 4$ subsets are floor-bound above 0.05 by construction; the $n = 5, 6$ results are genuinely underpowered (Section 6.4 estimates $n \approx 8$ –13 rungs needed for 80% power). No subset reaches $p < 0.05$; we therefore report drift floors, bootstrap intervals, and the explicit subset audit rather than a binary accept/reject verdict.
- **Label-permutation null — flag: null:label-pending.** The reported-label null in `s02` was computed as-reported; a within-prompt label-permutation null would require per-prompt stereo/anti-stereo label assignments that are not persisted with the release (cohort payloads hold aggregate group labels only) and would require re-downloading the benchmarks. We therefore rely on the *expert-index* permutation

null, which is marginal-preserving by construction and on-disk reproducible (s02: mean 0.31, sd 0.03, $p_{95} = 0.37$; only 0.5% of observed cohort pairs exceed p_{95}).

- **Top- k plumbing** — **flag:** `hooks:pending`. k (active experts) is read from router attributes at runtime (`hooks.py`); DBRX's N_A and any future config divergence may alter k/N assignment.
- **Exp5 benchmark set** — **flag:** `exp5:bench-vs-ladder`. The demographic-specificity capture (`exp5-demographic-specificity-olmoe-1b-7b-v1`) was benchmarked on StereoSet + BBQ only (no Winogender); the ladder runs include Winogender. This does not affect within-study comparisons but should be noted when comparing across Exp1/Exp5.

9 Conclusion

Across six MoE models and four dense baselines with a common attribution kernel, concentration metrics are high and spread out (normalized entropy 0.789–0.917; top-5 share 2–11%), with a monotone sparsity-to-concentration trend that is directionally consistent with H1 in every defensible subset—all-6 $\rho_H = +0.754$ ($p = 0.106$), paper-valid-4 $\rho_H = +0.949$ ($p = 0.167$), unique-sparsity-3 $\rho_H = +1.000$ ($p = 0.333$)—but that does not reach statistical significance at 0.05 on $n \leq 6$ ladder points. The robust signals are the dense-vs-MoE offset (a player-partition geometry effect) and the absence of demographic specialist structure. Expert-level pruning of social bias is not supported as a strategy by this data.

Take-away. Sparser routing does tend to concentrate bias more, but the effect is too small and the sample too thin to call it proven; more importantly, bias is spread across too many experts—and no demographic group leans on a distinct subset of them—for “just prune the biased experts” to be a workable fix.

Data & Code

- Runs, `result.json` summaries, and pair/player meta-data: `expt-bias-1/results/*/result.json`.
- Per-pair Shapley payloads (`per_pair_phi*.npy`, ≈ 565 MB across 15 captures — the original 8 plus 4 dense v1 baselines, the GPT-OSS-120B 5000-pair stability replication, and the two Exp8 same-mechanism LOO captures) exceed the code repository's file-size limits and are hosted on an external, anonymized data-sharing platform; a de-anonymized dataset link will be provided upon acceptance.
- Analysis scripts: `paper_figures_seaborn.py`, `s01_exp1_stability.py`, `s02_exp5_js.py`, `s03_h1_verdict.py`, `s04_bootstrap_cis.py`, `s05_exp5_bootstrap_js.py`, `s06_splithalf_extension.py`, `s07_lr_bootstrap.py`, and `audit_appendix.py` (all under `stats_analysis/scripts/`).
- Outputs and figures: `stats_analysis/outputs/`, `stats_analysis/figures/`.

Ethical Considerations

All benchmark prompts (StereoSet, BBQ, Winogender) are pre-existing, publicly released bias-evaluation datasets used only to measure, not to elicit or amplify, stereotyped completions; no new human-subjects data was collected. The routing-Shapley attribution measures where a contrastive log-probability shift localizes among experts, *not* whether a model is safe to deploy or how biased its outputs are in absolute terms; readers should not interpret a low top- k attribution share as evidence that pruning those experts would meaningfully reduce deployed bias; Section 5.6 shows ablation of the highest-attribution experts is a marginal, not exact, proxy for the causal effect, and is sometimes beaten by ablating high-routing-frequency experts instead. We do not release any model weights, fine-tunes, or jailbreak artifacts; all captures use publicly available, unmodified checkpoints.

Appendix A: Full Measurement Table and Drift Floors

Table 4 lists every model capture used in this paper. MoE ladder rows (OLMoE-1B-7B, Phi-3.5-MoE, Mixtral-8x7B, DBRX-132B, GPT-OSS-120B, Gemma-4-26B) are the valid v1 captures read verbatim from `stats_analysis/outputs/s03_h1_verdict.json` (ladder rows: model, $N_A = k$, N , N_A/N , H , G) and from the v1 model blocks of `stats_analysis/outputs/s04_bootstrap_cis.json` (`entropy[0..1]`, `gini[0..1]`, `t5[...]`, `t10[...]`); n_{pairs} comes from each run’s result.json. The Exp 5 OLMoE capture is read from `results/exp5-demographic-specificity-olmo-1b-7b-v1/result.json` (`concentration_metrics`). The dense baselines are the v1 per-pair captures used in Table 3, read from the `exp2-*v1/per_pair_phi.npy` payloads (1800/4000/1800/1800 pairs); the earlier v0 single-shot captures (150/400/400/400 pairs) are superseded. Bootstrap CIs for the dense rows come from the same block-bootstrap procedure as the MoE ladder (5000 resamples, seed 42, stratified by prompt group), reported alongside the MoE rows in `s04_bootstrap_cis.json`. GPT-OSS-120B shows its valid 2000-pair v1 capture; a 5000-pair replication (`exp1-concentration-gpt-oss-120b-5000`) confirms stability ($H = 0.8789$ vs. 0.8765 , within the CI). The k/N values are the verified N_A/N from `s03_h1_verdict.json` (Gemma corrected N_A : $960 \rightarrow 240$).

Drift floors. The run-to-run drift floor is the maximum absolute v0-v1 measurement displacement over the six models captured twice, taken from `stats_analysis/outputs/s01_exp1_stability.json` (per-model `dh/dg`; max absolute values: H 0.022997797740857084, G 0.0485411756982419). The stability module rounds these to the floors $\text{floor}_H = 0.0230$ and $\text{floor}_G = 0.0485$, as stated in the `final.rationale.drift` field of `s03_h1_verdict.json` (`‘floor.H 0.0230 / floor.G 0.0485’`). Table 4 therefore lists the per-model absolute drifts ($-\Delta H$, $-\Delta G$); the largest is Gemma’s $|\Delta H| = 0.022997797740857084$ (the floor itself) and OLMoE’s $|\Delta G| = 0.0485411756982419$ (the floor itself). Ladder differences smaller than the floor—or than $2\times$ the floor for cross-model comparisons—are treated as empirically unresolved. If a value is absent from the JSONs it is marked ‘—’ and never invented.

Localizability ratio and bias-gap-magnitude parity. A per-model *localizability ratio* $LR = \bar{H}_{\text{dense}}/H_{\text{MoE}}$ (mean dense entropy $\bar{H}_{\text{dense}} = 0.7043$ divided by each MoE rung’s H): OLMoE 0.802, Phi-3.5-MoE 0.801, Mixtral 0.768, DBRX 0.786, GPT-OSS-120B 0.804, Gemma-4-26B 0.893—every value is < 1 , i.e. dense is uniformly more concentrated, including on matched model families (OLMoE vs. its OLMo-7B dense sibling, $LR = 0.800$; Phi-3.5-MoE vs. Phi-3.5-Mini, $LR = 0.863$). With the same block bootstrap as Table 4 ($n_{\text{boot}} = 5000$, seed 42; the four dense- H draws are shared across rungs, so the six CIs are correlated and are reported for their per-rung coverage, not as independent evidence), the rung CIs are: OLMoE [0.782, 0.823], Phi-3.5-MoE [0.779, 0.821],

Mixtral [0.750, 0.790], DBRX [0.763, 0.804], GPT-OSS-120B [0.783, 0.824], Gemma-4-26B [0.865, 0.911]—every interval lies below 1—and the matched-family CIs are [0.763, 0.844] and [0.832, 0.884]. At the model level the split is complete: all four dense H values (max. 0.759) lie below all six MoE H values (min. 0.789); enumerating all $C(10, 4) = 210$ labelings of the ten model-level entropies under the permutation null, the observed configuration is the most extreme possible (two-sided $p = 0.0048$), and a six-of-six sign test on the rung LRs gives $p = 0.016$ (rungs treated as iid units; an upper bound given the shared dense mean). The dense/MoE split is therefore not a sampling artifact; we still report the drift floors above because LR inherits the same run-level variance as any single capture. Separately, the mean *absolute* contrastive bias-gap magnitude (from `bias_scores.mean.bias_gap` in each `result.json`, not a concentration metric) is 0.183 for the four dense baselines and 0.183 for the five non-null-bias MoE rungs, Cohen’s $d = 0.011$ (excl. Gemma). An exact model-level permutation test over all $C(9, 4) = 126$ labelings of the nine run-level means gives two-sided $p = 0.992$ (Welch t -test: $p = 0.986$)—statistically indistinguishable. Per-model values: OLMoE 0.244, Mixtral 0.200, DBRX 0.170, Phi-3.5-MoE 0.161, GPT-OSS-120B 0.141, Gemma-4-26B -0.142 (signed; null-bias routing noise, magnitude 0.142) vs. dense OLMo-7B 0.156, Llama-2-7B 0.173, Llama-3.1-8B 0.188, Phi-3.5-Mini 0.215. Including Gemma-4-26B’s signed value lowers the MoE-group mean to 0.129 (permutation $p = 0.66$ over $C(10, 4) = 210$ labelings; conclusion unchanged). Concentration geometry ($d_H = 3.91$, Section 5.4) and bias magnitude ($d_{\text{gap}} = 0.011$) are therefore orthogonal quantities, reinforcing the routing-structure-not-magnitude framing used throughout this paper.

Split-half shard agreement (all six ladder models). For Mixtral-8x7B and DBRX-132B, the two v1 captures originally submitted as two independent GPU shard jobs, each shard captured the pairs it was assigned by `pairs[shard_idx::num_shards]` (`scripts/run_bias_study.py`)—an *interleaved* even/odd split of the pair list, not a contiguous one. Recomputing concentration metrics independently on each shard (`stats_analysis/outputs/s01_exp1_stability.json` `shards` field): Mixtral shard 0 $H=0.9148$, $G=0.5236$; shard 1 $H=0.9173$, $G=0.5177$ ($|\Delta H|=0.0024$, $|\Delta G|=0.0058$); DBRX shard 0 $H=0.9040$, $G=0.5867$; shard 1 $H=0.9045$, $G=0.5831$ ($|\Delta H|=0.0005$, $|\Delta G|=0.0036$). The remaining four v1/replication captures (OLMoE, Phi-3.5-MoE, Gemma-4-26B, and the GPT-OSS-120B 5000-pair replication) were single GPU jobs with no shard split at capture time; we extend the same check *post hoc* by partitioning each stored `per_pair_phi.npy` array by row parity (even/odd pair index) and recomputing metrics on each half with the identical aggregation used throughout this paper (`stats_analysis/scripts/s06_splithalf_extension.py`, validated against Mixtral’s true shard numbers above to within 0.0004): OLMoE $|\Delta H|=0.0011$, $|\Delta G|=0.0027$; Phi-3.5-MoE $|\Delta H|=0.0079$, $|\Delta G|=0.0180$; Gemma-4-26B

Model	Family	N	k	k/N	Run dir	H	Gini	CL-H	CL-Gini	t_s	t_{exp}	$-\Delta H$	$-\Delta G$
OLMoE-1B-7B	olmo	1024	16	0.015625	exp1-concentration-olmoe-1b-7b-v1	0.8763468877475243	0.6472275458006027	0.877229232590607	0.8856719097568532	0.6300972550801518	0.6490112142692366	0.0916327653489081	0.5097878051421456
Phi-3.5-MoE	phi3.5moe	512	64	0.125	exp1-concentration-phi3.5-moe-v1	0.8765423790462808	0.639668261083971	0.8774773620432235	0.8917890948330988	0.6162873698172806	0.6430293910022759	0.0950471657642045	0.4693726921038138
Mixtral-8x7B	mixtral	256	64	0.25	exp1-concentration-mixtral-8x7b-v1	0.917431784769429	0.5164687423322485	0.913163877456115	0.934741324847850	0.4942736262170152	0.5286325505856332	0.1073384476079868	0.3568070327760832
DBRX-132B	dbxr	640	160	0.25	exp1-concentration-dbrx-v1	0.866428192510975	0.599019242351312	0.8665808302410779	0.909788288133642	0.5692323002143323	0.6031910619489534	0.0911109629767983	0.43573360397164
GPT-OSS-120B	gpt-oss	659	144	0.03125	exp1-concentration-gpt-oss-120b-v1	0.876490311375987	0.7273864977692972	0.876992237965315	0.8857548862669352	0.70986735996533	0.7307732155233386	0.0198254371016506	0.573122729723181
Gemma-4-26B [†]	gemma4moe	3840	240	0.0625	exp1-concentration-gemma4-26b-v1	0.7887735094946339	0.8551293373691959	0.7912880207701668	0.8018981269787994	0.841279214151506	0.8525262679470473	0.050593415428668	0.766853661303078
OLMoE-7B (dense)*	olmo	32	—	—	exp2-dense-baseline-olmo-7b-v1	—	0.702576	—	0.719123	—	0.739277014834522	—	0.749066
Phi-3.5-Mini (dense)*	phi3.5	32	—	—	exp2-dense-baseline-phi3.5-mini-v1	—	0.759191	—	0.764503	—	0.7774305357401991	—	0.78709796072026446
Llama-2-7B (dense)*	llama	32	—	—	exp2-dense-croscrocheck-llama-2-7b-v1	—	0.696892	—	0.654052	—	0.658746217347756	—	0.72958028440434
Llama-3.1-8B (dense)*	llama	32	—	—	exp2-dense-croscrocheck-llama-3.1-8b-v1	—	0.6587063	—	0.707863	—	0.630231291108987	—	0.7012513439006255
OLMoE-1B-7B (Exp 5)	olmo	1024	16	0.015625	exp1-demographic-specificity-olmoe-1b-7b-v1	0.878624280562932	0.6456658511909388	—	—	—	—	0.09276401412301853	0.50876675632717

Table 4: Full measurement table. MoE rows: v1 captures from s03_h1 verdict.json ladder and s04_bootstrap_cis.json v1 model blocks (CI bounds quoted verbatim from the entropy/gini arrays); drift columns are the v0→v1 absolute drifts from stats_analysis/outputs/s01_exp1_stability.json (dh/dg). Dense rows: v1 per-pair captures from the exp2-*v1 runs as reported in Table 3; CIs come from the same s04_bootstrap_cis.json block bootstrap as the MoE ladder (no drift column: the s01_exp1_stability.json drift-floor analysis covers only the six-model MoE ladder, not the dense baselines, so no v0→v1 comparison is reported for dense rows). Exp 5 row: pooled OLMoE concentration from exp5-demographic-specificity-olmoe-1b-7b-v1/result.json (it is not among the s04 bootstrap blocks, so its CI cells are marked ‘—’). [†]Gemma-4-26B is a null-bias model (mean bias gap ≈ 0); see Section 8. *Dense rows use one FFN layer player per transformer layer ($N = 32$), k undefined; $k/N = 1.0$ as in the dense-group row of Table 1.

$|\Delta H| = 0.0016$, $|\Delta G| = 0.0015$; GPT-OSS-120B (5000-pair) $|\Delta H| = 0.0022$, $|\Delta G| = 0.0058$. All six ladder models now have a split-half check (two by independent GPU capture, four post hoc) and agree to $|\Delta H| \leq 0.008$ and $|\Delta G| \leq 0.018$ on independently-sampled prompt-pair halves—an internal-consistency check distinct from the v0→v1 rerun-drift floor above (which compares different pair counts, not disjoint samples at the same count).

Appendix B: Causal Ablation and Proxy-Agreement Robustness

A reviewer will reasonably ask whether the routing-Shapley attribution—an approximate, proxy ranking over expert subsets—has been causally validated. The study catalog (expt-bias-1/study-catalog.txt, sections “Reviewer-Prioritized Extensions”) defines four experiments that speak to exactly this question. Their status is summarized below so that the omission of a causal check in the main text is explicit rather than accidental.

Experiment 4/6 — causal expert ablation (DONE for 6 models; GPT-OSS-120B not yet landed). Experiment 4 (‘Independent cross-check (robustness)’) validates Shapley rankings against expert-ablation disparity drops, and Experiment 6 extends it ladder-wide: ablating experts in ϕ -ranked order and measuring how fast the contrastive bias gap drops, compared against two controls—random expert ablation and ablation of the highest-routing-frequency (high-routing) experts. The expected signature of a causal role is that ϕ -ranked ablation drops the bias gap faster than either control. **Status: DONE for six models.** Ablation curves exist on disk at results/{exp1-concentration-olmoe-1b-7b, exp1-concentration-mixtral-8x7b, exp1-concentration-phi3.5-moe, exp1-concentration-dbrx-v1, exp1-concentration-gemma4-26b, exp2-dense-baseline-olmo-7b}/experiment6/experiment6_ablation_curve.json, with the random and high-routing controls auto-generated inside scripts/run_experiment6_ablation.py (shapley.compute_ablation_curve, k -step schedule at 1–10 and 10%, 20%, 30%, 50%, 75%, 100% of players; each curve

starts from an unablated baseline point $k=0$). Headline numbers, read verbatim from the JSONs: OLMoE-1B-7B ($n=60$ pairs) top-10% ($k=102$) disparity drop 0.567 (ϕ), -0.064 (random), 0.374 (high-routing); Mixtral-8x7B ($n=30$) top-10% ($k=26$) -0.121 , -0.005 , 0.663; Phi-3.5-MoE ($n=30$) top-10% ($k=51$) 0.302, -0.364 , -0.219 ; DBRX-132B ($n=60$) top-10% ($k=64$ of 640) -0.101 , -0.110 , -0.392 (all three orders increase the gap at this fraction), 50% ($k=320$) $+0.266$, $+0.996$, $+1.186$ (ϕ trails both controls badly); Gemma-4-26B ($n=60$) is uninterpretable—its baseline bias gap is 0.0182 (null-bias flag), so disparity_drop at 10% ($k=384$) is $+4.12$ (ϕ), $+6.95$ (random), $+8.96$ (high-routing)—dividing by a near-zero denominator produces 4–9× swings that carry no causal information; OLMoE-7B dense ($n=60$, $N=32$ layer players) has no high-routing control (no router) and overshoots at $k=1$ (drop 1.083), reflecting its 32-player partition. The full-curve analysis, including the reversal on DBRX, is reported in Section 5.6. GPT-OSS-120B’s ladder run (job 5575800) was running at time of writing when the cluster entered a scheduled maintenance window (2026-08-11 06:00–2026-08-13 23:59); its final state is not yet known.

Experiment 7 — proxy-vs-exact-Shapley agreement (DONE; null result on all three tested models). The routing-contrast kernel is an approximation of the exact Shapley value; Experiment 7 quantifies the gap by computing the Spearman ρ between the routing-contrast ranking and the exact-Shapley ranking, over $n=20$ pairs and the first/last MoE layer, on every model where exact computation is tractable ($K=2$ –16 experts active per layer). Results exist on disk for three models: Mixtral (results/exp1-concentration-mixtral-8x7b/experiment7/experiment7_proxy_agreement.json, mean_spearman_rho -0.085 layer 0 sd 0.373, $+0.058$ layer 31 sd 0.445), OLMoE (results/exp1-concentration-olmoe-1b-7b/experiment7/experiment7_proxy_agreement.json, -0.024 layer 0 sd 0.287, $+0.235$ layer 15 sd 0.395), and Phi-3.5-MoE (results/exp1-concentration-phi3.5-moe/experiment7/experiment7_proxy_agreement.json, -0.084 layer 0 sd 0.341, $+0.076$ layer 31 sd 0.311). **Status:**

DONE for three models. Every per-pair agreement estimate across all three models and both layers is statistically indistinguishable from zero ($|\rho| \leq 0.24$, sd 0.29–0.45 on $n=20$ pairs) — the proxy ranking does not reproduce exact-Shapley ordering within this pair budget on any tested model, consistent with the noisy small- k regime of the causal curves in Table 4 and Section 5.6.

Experiment 8 — same-mechanism comparison (DONE with per-pair CIs for both models). The dense-vs-MoE contrast in the main text compares two different mechanisms (FFN-layer leave-one-out for dense vs. expert-subset routing-contrast for MoE). Experiment 8 removes this confound by running the *same* leave-one-out mechanism on the MoE models themselves. Both result directories exist: `results/exp8-1100-olmoe-1b-7b/result.json` (OLMoE, $n_{pairs} = 100$, $N = 16$ FFN-layer players, $H = 0.7361$, Gini = 0.6239) and `results/exp8-1100-phi3.5-moe/result.json` (Phi-3.5-MoE, $n_{pairs} = 50$, $N = 32$, $H = 0.8993$, Gini = 0.4514). The catalog’s key question—does layer-level LOO entropy on MoE stay high (≈ 0.90) or drop toward the dense band 0.66–0.76?—has a partial answer: OLMoE’s $H_{100} = 0.736$ lands inside the dense band, while Phi-3.5-MoE’s $H_{100} = 0.899$ stays near the MoE range. Per-pair captures now exist for both models, supporting 95% stratified block-bootstrap CIs ($n_{boot} = 5000$, seed 42): Phi-3.5-MoE $H \in [0.842, 0.941]$, Gini $\in [0.348, 0.544]$ — consistent with the MoE range and excluding the dense band’s upper edge; OLMoE $H \in [0.692, 0.921]$, Gini $\in [0.355, 0.657]$ — a much wider interval (only $n = 100$ dense-FFN-layer pairs, no stratification benefit from a larger ladder-style capture) that overlaps both the dense band and the low end of the MoE range, so OLMoE’s same-mechanism entropy is directionally dense-like but not statistically distinguishable from the MoE range at this pair count. This 2-model comparison is itself only directional; configs extending the same-mechanism check to Mixtral, DBRX, GPT-OSS-120B, and Gemma-4-26B are written and ready to submit (`configs/study.\{mixtral-8x7b,dbxr,gpt-oss-120b,gemma4-26b\}.1100.yaml`) but not yet run. Extending the LOO mechanism to Gemma-4-26B required a code change first—its decoder layer sums two parallel FFN branches (an always-on dense branch plus the routed experts) that a naive single-module ablation would only partially zero; the compound branch-ablation path is implemented in `shapley.compute_dense_layer_contrast` and the four extension configs are ready to submit once the cluster maintenance window closes. **Status: DONE with bootstrap CIs for both OLMoE and Phi-3.5-MoE; ladder extension to 4 more models queued.**

Experiment 3 — expert-pair synergy / collectivity check (DONE for 4 models; DBRX and GPT-OSS-120B not yet landed). Marginal Shapley values ϕ_i assume experts contribute independently; Experiment 3 tests this by computing pairwise Shapley *interaction* values Φ_{ij} on the first and last routed MoE layer ($n=20$ pairs, `scripts/run_experiment3_interactions.py`, `shapley.compute_shapley_interactions_for_pair`) and

reporting the synergy fraction $\sum |\Phi_{ij}| / (\sum |\phi_i| + \sum |\Phi_{ij}|)$. **Status: DONE for four models.** OLMoE-1B-7B: 0.705 (layer 0) / 0.281 (layer 15, last); Phi-3.5-MoE: 0.741 / 0.200 (layer 31, last); Mixtral-8x7B: 0.716 / 0.503 (layer 31, last); Gemma-4-26B: 0.747 / 0.183 (layer 29, last). All four models show the same shape: synergy dominates (0.70–0.75 of attribution mass) in the first routed layer and falls by roughly half at the last layer, but never disappears. This means the ϕ -ranking used by the causal ablation (Exp. 4/6, above) discards 18–75% of the true attribution mass depending on model and layer, a plausible mechanistic explanation for the ablation reversals reported above (DBRX, and Mixtral’s high-routing win): where synergy is high, marginal ϕ -ranking is a poor proxy for which experts actually carry the removable bias. DBRX and GPT-OSS-120B collectivity runs (jobs 5575799, 5575791) were running at time of writing when the cluster entered the same scheduled maintenance window (2026-08-11 06:00–2026-08-13 23:59); their final state is not yet known.

Net status for reviewers. Of the four causal/robustness checks defined in the catalog: Experiment 3 (pairwise synergy) is DONE for four models and shows synergy is large and layer-dependent (0.18–0.75 of attribution mass); Experiment 4/6 (causal ablation) is DONE for six models (OLMoE, Mixtral, Phi-3.5-MoE, DBRX, Gemma-4-26B, and the OLMo-7B dense control; Section 5.6) with GPT-OSS-120B not yet landed; Experiment 7 (proxy agreement) is DONE and reports a null result on all three tested models (per-pair $\rho \approx 0$ on Mixtral, OLMoE, and Phi-3.5-MoE); and Experiment 8 (same-mechanism) is DONE with bootstrap CIs for both OLMoE and Phi-3.5-MoE. The combined causal evidence is weaker than a single-model reading would suggest: on OLMoE and Phi-3.5-MoE the ϕ -ranked ablation drops the bias gap faster than random ablation; on Mixtral the high-routing-frequency control outperforms ϕ ; on DBRX ϕ -ranking is outright *worse* than both controls at the half-ladder point; and Gemma-4-26B’s near-zero baseline bias gap makes its ablation curve uninterpretable. No model supports a small fixable expert set, and the Exp. 3 synergy finding gives a mechanistic reason why: a purely marginal ranking is a partial proxy whenever pairwise interaction carries a large share of the attribution mass, which is every model we tested to varying degrees.

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